ECAI 2025
I. Lynce et al. (Eds.)
© 2025 The Authors.
This article is published online with Open Access by IOS Press and distributed under the terms of the Creative Commons Attribution Non-Commercial License 4.0 (CC BY-NC 4.0).
doi:10.3233/FAIA250797

Scribble-Based Weakly Supervised Camouflaged Object Detection via SAM-Guided Feature Correlation Transformer

Zi-Jie Wu[a], Rongrong Gao[b] and Tian-Zhu Xiang[c,*]

[a]School of Computer Science and Engineering, Guangxi Normal University, Guilin, China
[b]Department of Computer Science and Engineering, HKUST, Hong Kong, China
[c]Space42, Abu Dhabi, UAE

Abstract. Weakly Supervised Camouflaged Object Detection (WSCOD) aims to locate camouflaged objects with only sparse supervision, thereby substantially reducing the reliance on costly pixel-level annotations. This task poses two major challenges: limited supervision arising from sparse annotations (*e.g.*, scribbles), and weak discriminability due to the inherent high visual similarity between camouflaged objects and their surroundings. To tackle these challenges, this paper proposes a novel multi-scale feature correlation transformer guided by the Segment Anything Model (SAM) for scribble-based WSCOD. Specifically, we introduce a cross-scale correlation module built upon Transformers, which exploits enriched cross-attention mechanisms to capture long-range global correlations and multi-scale discriminative cues, enabling accurate segmentation of camouflaged objects. In addition, we develop a SAM-based pseudo-label generation module that leverages sparse annotations as prompts to produce high-quality object masks, thereby enhancing supervision. Extensive experiments on three challenging datasets demonstrate that our proposed method consistently and significantly surpasses existing state-of-the-art approaches for scribble-based WSCOD. The code will be available at: http://github.com/farewellIamLoser/FCT-SAM-WSCOD.

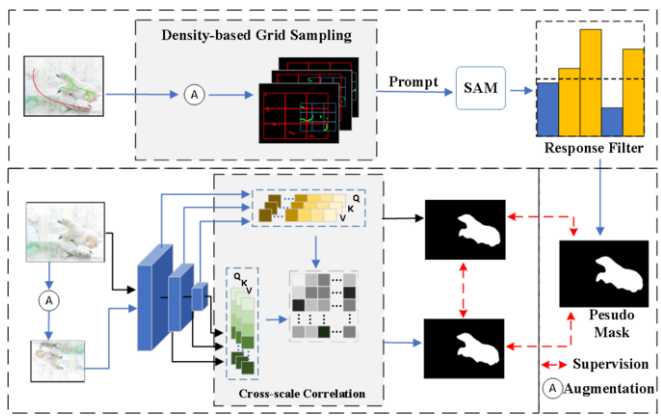


Figure 1: Illustration of the proposed method, including SAM-based pseudo label generation and multi-scale feature correlation transformer. The proposed method adopts sparse annotations as prompts for SAM to generate high-quality pseudo labels for model training. To enhance the quality of pseudo masks, we introduce a set of strategies, including a density-based grid sampling for prompt enhancement and a label-response filtering mechanism.

1 Introduction

Camouflaged object detection (COD) refers to identifying and segmenting objects that are intentionally designed or naturally evolved to blend in with their surroundings [12, 52]. Gaining growing attention within the computer vision community, camouflaged object detection holds considerable promise for a variety of applications, including species discovery [39], medical image segmentation [27, 28], agricultural pest detection [1], and industrial defect detection [13].

In recent years, there has been a considerable amount of research for deep learning-based camouflaged object detection, such as biologically inspired approaches [12, 35, 37], coarse-to-fine learning methods [18, 19], multi-task joint learning methods [26, 32, 41], confidence-aware learning methods [26, 54], and multi-source information guidance, *e.g.*, frequency [57] and depth [45]. Although great progress has been achieved, obviously, these methods heavily depend on large-scale datasets with high-quality pixel-wise annotations [8, 48]. As we know, it is time-consuming and labor-intensive to annotate a pixel-wise camouflaged object from its highly similar background. To this end, Weakly Supervised Camouflaged Object Detection (WSCOD) is proposed to detect camouflaged objects in complex scenes using only weak supervision (*e.g.*, sparse annotations) rather than precise pixel-level labels. This setting imposes greater demands on a model's ability to learn from limited supervision while maintaining robust visual detection performance. By alleviating the need for costly pixel-wise annotations, weak supervision greatly reduces data annotation overhead, making WSCOD more practical and scalable for real-world applications where large-scale pixel-wise annotations are expensive and time-consuming to obtain. CRNet [17] introduces the first weakly-supervised COD framework trained with scribble annotations. To enhance weak supervision, [16] explores the use of Segment Anything Model (SAM) [23] to generate dense pseudo-labels as supervision from sparse annotations.

Nevertheless, these methods still show a significant performance gap compared to fully supervised COD methods. Based on our observation, there are two main limitations. (1) The sparse annotation only provides limited supervision for model training, thus greatly restricting the discrimination capacity of models. Although [16] has shed

* Corresponding Author. Email: tianzhu.xiang19@gmail.com

some light, there is still significant room for improvement. It is desirable to explore how to obtain accurate and effective dense pseudo labels from sparse labels. (2) The intrinsic high similarity between camouflaged objects and their backgrounds makes it far more challenging to identify camouflaged objects. How to effectively explore and aggregate subtle camouflaged cues remains a challenge.

To this end, in this paper, we propose a novel Segment Anything Model (SAM) guided multi-scale Feature Correlation Transformer, termed *FCT-SAM*, shown in Fig. 1, for the challenging scribble-based weakly-supervised camouflaged object detection. The proposed method designs a novel SAM-based pseudo-label generation strategy to strongly guide the global context learning of camouflaged features, which explores and aggregates effective discriminative camouflage cues from different scales for accurate object segmentation. Specifically, to improve the weak supervision of sparse annotations, we introduce the recently proposed segment anything model and explore the use of sparse annotations as prompts to generate dense segmentation masks (pseudo labels) for model training. It is well known that direct use of SAM for camouflaged scenes does not always yield satisfactory results [21]. To alleviate the issues aforementioned, we design two strategies, that is, prompt enhancement and label response filter, to filter out the optimal masks and enhance the label quality for more reliable supervision. Moreover, to enhance the exploration of camouflage cues, we present a novel cross-scale correlation (CSC) module, which leverages enriched cross-attention mechanisms to model global context and capture long-range correlations from multi-scale features for strengthening feature representation. The proposed module can effectively excavate and aggregate discriminative camouflage cues for accurate segmentation. In summary, our contributions are as follows.

- We introduce a novel SAM-based pseudo-label generation module, which uses sparse labels as prompts for SAM to generate high-quality dense segmentation masks by the designed prompt enhancement and label response filter strategies. The generated pseudo-labels offer reliable guidance for training models in weakly-supervised camouflaged object segmentation.
- We propose a cross-scale correlation (CSC) module to extract and aggregate subtle camouflage cues from features at different scales for WSCOD. It adopts enriched cross-attention operations to capture discriminative cues from global context learning, significantly improving the representation of camouflaged features.
- Extensive experiments show that our proposed method achieves superior performance over the existing state-of-the-art approaches by a large margin on three COD benchmark datasets, for scribble-based weakly supervised camouflaged object detection.

2 Related Work

2.1 Camouflaged Object Detection

The rapid advancement of deep learning has given rise to numerous fully supervised models within the realm of camouflaged object detection (COD) [13]. The existing COD methods can be broadly categorized into the following categories, including bio-inspired approaches, coarse-to-fine learning, multi-task joint learning, and multi-source information guidance.

The bio-inspired approaches mimic the human visual system, such as search and identification [12], positioning and focus [35], and zoom in and out [37]. The coarse-to-fine learning methods perform COD by exploring camouflage features at different scales and aggregating global and local camouflaged cues, such as FSPNet [19] and HitNet [18]. The multi-task joint learning facilitates object segmentation through joint modeling of multiple related tasks, such as boundary detection [41], saliency detection [26], and ranking [32]. The multi-source information guidance introduces multi-modal cues to guide camouflage feature learning, such as frequency domain [57], depth [45], and temporal cues [38]. More recently, several approaches have integrated generative models into COD, with diffusion-based methods showing notable potential in enhancing detection performance, such as [6, 7]. Nevertheless, these significant achievements have been made in a fully supervised manner that relies on a large amount of training data with pixel-level labels [25, 44].

Compared to fully supervised COD, its weakly supervised counterpart, relying on sparse annotations such as points, boxes, or scribbles, is even more challenging and has not been relatively underexplored. CRNet [17] introduces the first weakly-supervised COD model using scribble annotations, which designs a group of loss functions to supervise the convolution model for accurate predictions. Note that the scribble annotations used in this work are sourced from the public WSCOD dataset provided by CRNet. WS-SAM [16] proposes a multi-scale feature grouping module to mine discriminative cues at different granularities, with enhanced supervision from SAM-generated pseudo labels. SAM-COD [3] employs a prompt adapter to handle scribbles as prompts based on SAM to improve the quality of masks. According to our observations, the limitation of sparse annotations and the intrinsic high similarity between objects and backgrounds remain challenges that restrict the discrimination capacity and diminish the segmentation performance. In this paper, we propose a novel multi-scale feature correlation transformer to explore subtle discriminative camouflage cues from global context learning, guided by our designed SAM-based pseudo-label generation strategy, significantly enhancing the WSCOD performance.

2.2 Segment Anything Model in COD

Segment Anything Model (SAM) [23] is a recently proposed advanced vision foundation model designed for general object segmentation. SAM is a promptable segmentation model that produces object masks in any given image from input prompts such as points or boxes. Its remarkably exceptional performance has rapidly propelled its extensive application across various computer vision tasks.

However, recent studies have found that SAM faces challenges and performs poorly when segmenting objects in certain complex or ambiguous scenes, such as camouflaged object segmentation [21]. To mitigate issues, SAM-adapter [5] introduces adapters for efficient tuning to combine domain-specific information or visual prompts into a segmentation model. Unlike fine-tuning with full pixel-wise labels, SAM is also adopted as a pseudo-label generator for weakly supervised semantic segmentation, which explores the potential of "prompt to masks" [4]. In particular, recent WS-SAM [16] and SAM-COD [3] utilize sparse labels as prompts for SAM to generate pseudo-object masks for WSCOD. In contrast to these works, we explore prompt enhancement for weakly supervised labels and a response filter strategy to generate more precise and reliable masks as strong supervision signals for performance improvement.

3 Proposed Method

3.1 Overall Architecture

The overall framework of our proposed method is illustrated in Fig. 1, which consists of two components: SAM-based pseudo-label

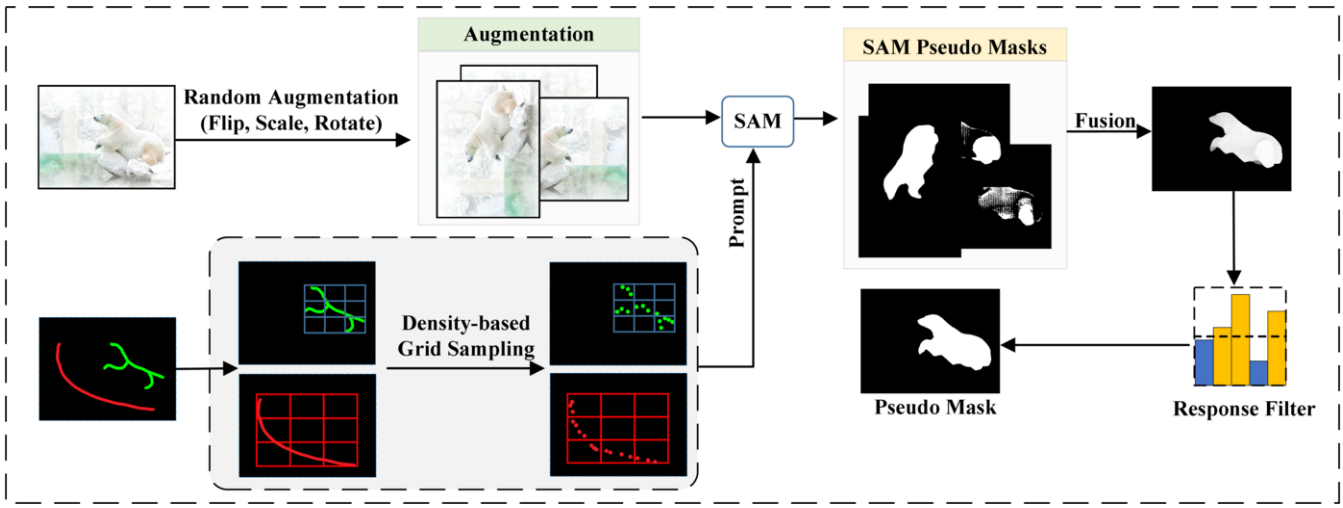


Figure 2: The proposed SAM-based pseudo-label generation. The method mainly contains two operations: prompt enhancement based on density-based grid sampling and label response filter, to enhance the quality of pseudo labels for more reliable supervision.

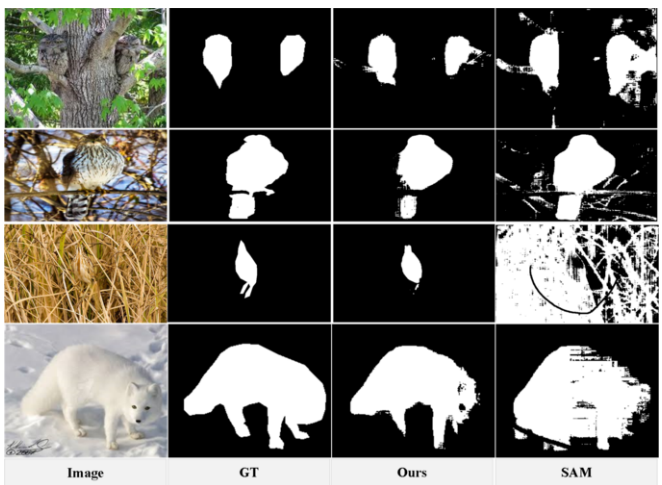


Figure 3: Visual comparisons of our proposed method and the SAM method for pseudo mask generation.

generation and multi-scale feature correlation transformer. The former, shown in Fig. 2, is designed to generate high-quality pseudo labels for model training using SAM. Specifically, we first adopt a density-based grid sampling strategy to enhance the prompts from scribble annotations, then feed them into SAM to produce multiple masks, which are then fused to generate the final pseudo labels via the response filter strategy. The latter, shown in Fig. 4, is an end-to-end transformer framework for camouflaged object segmentation. The proposed transformer accepts two kinds of inputs: the original image and the corresponding augmented images.

In the proposed model, the transformer encoder extracts multi-scale features from the inputs, which are utilized for camouflage feature learning in the designed cross-scale correlation module. The enriched cross-attention operations are employed to explore the global correlations and aggregate discriminative camouflage cues across different scales. The model ultimately produces two types of predictions from two inputs. Self-supervision (between two types of predictions) and pseudo-label supervision (between predictions and pseudo-labels) are applied separately for model training.

3.2 SAM-based Pseudo Label Generation

SAM [23] has demonstrated remarkable performance in generating high-quality object masks in an image, a feat attributed to its extensive training on a large-scale dataset comprising over one billion object masks. However, recent studies found that SAM still struggles in some confusing scenes [21], such as concealed scenes where objects are highly similar to their background, as shown in Fig. 3.

SAM is a promptable segmentation model that requires input prompts to generate masks. We observe that SAM has a limitation: when there are too many input prompts, the prompt points may affect the model's judgment, causing it to focus excessively on certain areas or ignore background prompt information. To address these challenges, this paper proposes a series of strategies based on the SAM model, including prompt enhancement and label response filter, as shown in Fig. 2. Through these operations, SAM can generate high-quality and reliable pseudo-labels.

Prompt Enhancement. The input camouflaged images and corresponding scribble labels are represented as $D = \{x_i,\ y_i\}_{i=1}^{n}$, where $\{x_i\}_{i=1}^{n}$ is the input image and $\{y_i\}_{i=1}^{n}$ is the scribble label. To improve prompt generation, we apply flipping, rotation, and scaling operations to the input image for image augmentation. By performing $\mathcal{K}$ rounds of random augmentation, we obtain multiple augmented images. For each augmented image, we sample $\mathcal{P}$ discrete pixel points $L = \{fg,\ bg\}$ in foreground and background scribbles as point prompts. More details about the sampling data can be found in Tab. 4.

To sample these points, we design a *density-based grid sampling* to sample a sufficient number of point prompts from various locations across the scribbles. The density-based grid sampling will find the leftmost, rightmost, highest, and lowest points of the scribble label based on their positions. A rectangle is then formed using these four points, and the rectangle is evenly divided into a $m \times m$ grid. In this paper, $m = 3$. Assuming the number of pixel labels as N_s, we divide the total number of label pixels within each grid GN_i by N_s, thus obtaining the density D_i for each grid, represented as:

$$D_i = GN_i / N_s. \tag{1}$$

Based on the predefined number of sampling points $\mathcal{P}$, we calculate the number of randomly sampled points N_i for each grid according

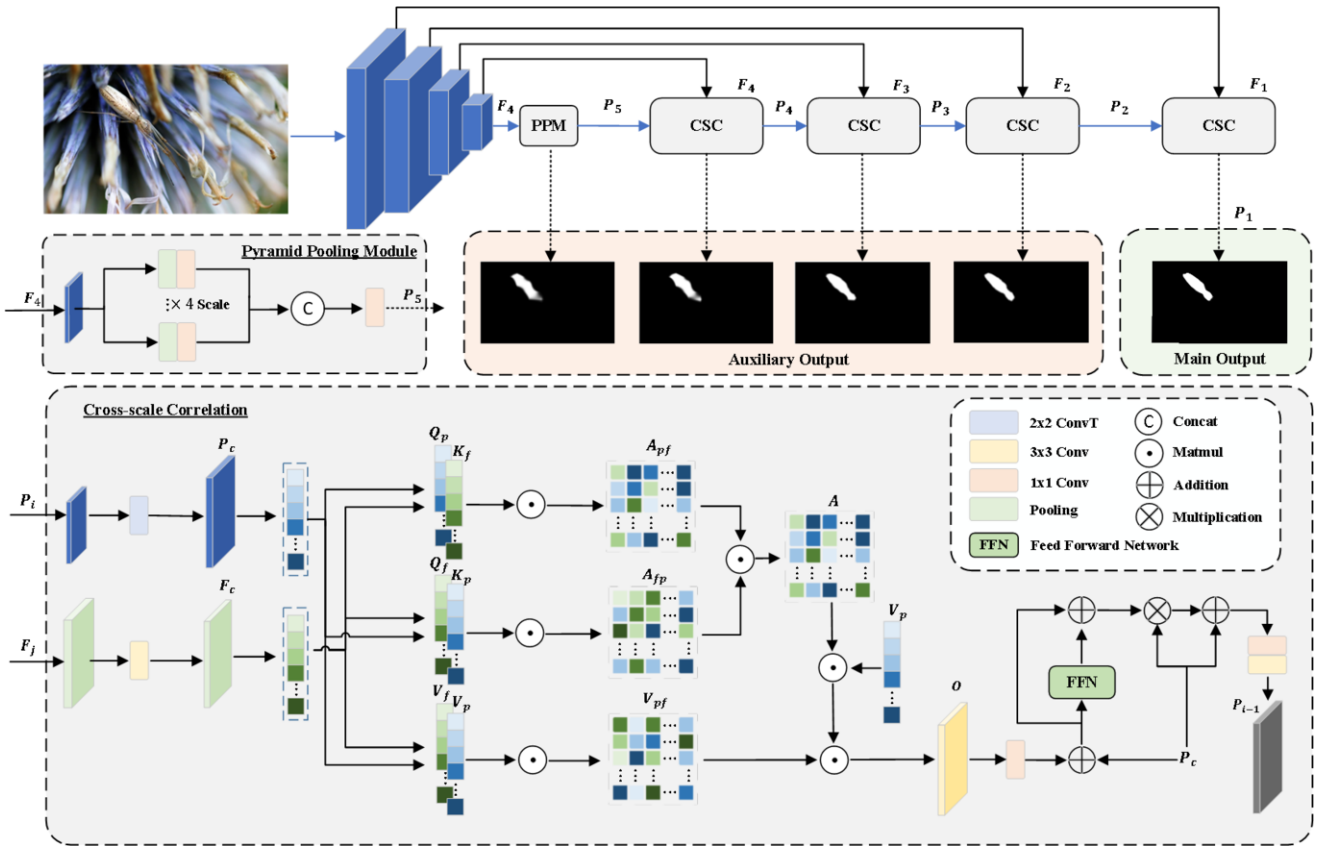


Figure 4: Overview of our proposed multi-scale feature correlation transformer. Four features at different scales ($\{F_i\}_{i=1}^{4}$) are extracted from the transformer encoder and subsequently fed into the designed cross-scale correlation (CSC) module for feature decoding and mask predictions. The last encoder feature F_4 is enhanced through a pyramid pooling module (PPM) before being fed into the CSC module. "ConvT" is transposed convolution.

to its density value D_i, denoted as:

$$N_i = \mathcal{P} \times D_i. \tag{2}$$

Next, within each grid cell, the N_i points are randomly sampled. The greater the density, the proportionally more sampling points. Based on this rule, we randomly sampled point prompts in each grid on the foreground ($\{fg_i\}_{i=1}^{N_{fg}}$) and background ($\{bg_i\}_{i=1}^{N_{bg}}$) scribbles, respectively. Through multiple iterative samplings, we obtain diverse SAM-generated masks M, which can be denoted as:

$$M_i^{k_e \times k_a} = \mathrm{SAM}(x_i^{k_a}, fg_i, bg_i), \tag{3}$$

where k_a is the number of image augmentation for x_i and k_e is the number of iteration sampling. Therefore, we obtain the initial SAM-generated object pseudo-mask set. As shown in Fig. 3, randomly sampled prompt points often result in suboptimal mask predictions due to their limited representativeness. Our method enforces a more uniform distribution across the entire scribble, improving prompt quality for the target region and boosting mask accuracy.

Label Response Filter. The generated pseudo-labels are not always reliable. Therefore, we design a label response filter to filter out the unreliable response pixels and improve pseudo-label robustness. Specifically, we first sum up all generated label responses to produce a pixel-wise response occurrence frequency map $\mathcal{M}$, denoted as:

$$\mathcal{M} = \frac{1}{|k_e \times k_a|} \sum_{i=0}^{k_e \times k_a} M_i. \tag{4}$$

Then we calculate the average response occurrence frequency μ:

$$\mu = \frac{1}{|S|} \sum_{(i,j) \in S} \mathcal{M}_{(i,\,j)}, \tag{5}$$

where S is the set of response pixels on the pseudo mask. $|S|$ is the total number of pixels of this type. Next, a threshold strategy is adopted to discard low-confidence responses, defined as

$$\tilde{\mathcal{M}}_{(i,\,j)} = \begin{cases} 0, & \mathcal{M}_{(i,\,j)} < \frac{\mu}{\tau} \\ 1, & \mathcal{M}_{(i,\,j)} \geq \frac{\mu}{\tau} \end{cases}, \tag{6}$$

where τ is the balance factor, and we set $\tau = 4$ in our work. $\tilde{\mathcal{M}}$ is the final generated pseudo label for model supervision.

3.3 *Multi-scale Feature Correlation Transformer*

To effectively capture camouflage cues, we design an end-to-end multi-scale feature correlation transformer. It utilizes the enriched cross-attention mechanisms of the transformer to mine global context cues of the camouflaged object, and then progressively aggregates these discriminative features scale by scale, for accurate segmentation of camouflaged objects. As shown in Fig. 1, the proposed model receives two kinds of input separately, that is, the normal image and the augmented image. Then the model produces two sets of corresponding mask predictions at different scales ($\{\{Z_i\}_{i=1}^{5}$ and $\{\tilde{Z}_i\}_{i=1}^{5}$). $\{Z_i\}_{i=2}^{5}$ and $\{\tilde{Z}_i\}_{i=2}^{5}$ are the side outputs, and Z_1 and $\tilde{Z}_1$ are the final main output. To supervise the model training, two kinds

of loss functions are adopted: (a) the structure consistency loss, and (b) the pseudo-label supervision for model training. The former can supervise the mask predictions based on two kinds of input images to maintain consistency over different images.

The entire model architecture is illustrated in Fig. 4. Let $I \in \mathbb{R}^{H\times W\times 3}$ denotes the input image. The PVT-v2 [43] is adopted as the transformer encoder to encode the input images into four features at different scales $\{F_i\}_{i=1}^{4}$. These encoded features are fed into the designed multi-scale correlation decoder to excavate and aggregate discriminative camouflage features for mask prediction. For feature decoding, a pyramid pooling module [55] is first adopted to enhance the feature representation for the last layer encoder output F_4 and obtain the enhanced feature P_5. Specifically, four pooling operations of different sizes are applied to obtain multi-scale features, followed by a standard 1×1 convolution block. With concatenation and convolution operations, we get the output feature P_5.

In our decoder, two features of adjacent scales F_j and P_i are fed into the designed cross-scale correlation (CSC) module for feature exploration and progressive aggregation. The output feature P_{i-1} of CSC will then be fed into the next CSC module together with the encoder feature F_{j-1} of the previous layer. The CSC module is cascaded until the aggregation of first-layer encoder features. The process can be denoted as:

$$P_{i-1} = \mathrm{CSC}\left(P_i, F_j\right), i \in \{5, ..., 2\}, j \in \{4, ..., 1\} \tag{7}$$

where P_1 is the final output for object mask prediction.

Cross-scale Correlation. To effectively mine and aggregate subtle discriminative camouflage cues from multiple scales, we design a cross-scale correlation (CSC) module, which utilizes enriched cross-attention operations to fully capture long-range pixel correlations from features at different scales and model global connectivity.

As shown in Fig. 4, the CSC module receives two feature inputs: the output feature P_i of the previous CSC module and the backbone feature F_j of the previous scale. To align the two feature sizes, a transposed convolution and a standard convolution are used to get P_c and F_c, respectively. Note that P_5 does not require transposed convolution before being fed into the CSC module. To enhance feature representation, the proposed CSC then applies three separate 1×1 convolutions followed by a 3×3 depth-wise convolution to explore the local context, thus generating query (Q), key (K) and value (V) projections for cross-attention computation. Similar to the conventional multi-head self-attention in the transformer, Q, K, V are also extended to a multi-head version for learning separate attention maps in parallel. Therefore, we obtain two sets of features Q_p, K_p, V_p and Q_f, K_f, V_f, respectively. Next, we reshape query and key projections and use matrix multiplication for cross-interaction to generate correlation attention maps, which encode the global context implicitly. This is defined as:

$$\begin{aligned} \mathcal{A}_{pf} &= \mathrm{Softmax}\left(Q_p^T K_f/\alpha\right) \\ \mathcal{A}_{fp} &= \mathrm{Softmax}\left(Q_f^T K_p/\alpha\right) \end{aligned}, \tag{8}$$

where α is a learnable scaling parameter to control the magnitude of the matrix multiplication. These two attention maps then interact to generate the final correlation attention map $\mathcal{A}$, that is, $\mathcal{A} = \mathcal{A}_{pf} \cdot \mathcal{A}_{fp}$. Before the attention operation, the two value projections V_p and V_f also interact with each other, yielding $V_{pf} = V_p^T \cdot V_f$. Then the attention map is assigned to the value projections to get the fully interactive feature O, denoted as:

$$O = (\mathcal{A} \cdot V_p) \cdot V_{pf}. \tag{9}$$

Next, a feed-forward network (FFN) is adopted to generate the output feature P_{i-1} for the next CSC module, which is formulated as:

$$\begin{cases} O^{'} & = W_1(O) \oplus P_c \\ P_{i-1} & = W_3(W_1((\mathrm{FFN}(O^{'}) + O^{'}) \otimes P_c \oplus P_c)), \end{cases} \tag{10}$$

where W_1 is the 1×1 convolution, W_3 is the 3×3 convolution, $\oplus, \otimes$ are element-wise addition and multiplication, respectively. For each feature $\{P_i\}_{i=1}^{5}$, a 1×1 convolution is adopted, followed by a Sigmoid function, to generate each mask prediction.

Loss Function. Our model utilizes three kinds of loss functions to supervise the prediction outputs: $\{Z_i\}_{i=1}^{5}$ and $\{\tilde{Z}_i\}_{i=1}^{5}$. Based on SAM-generated pseudo labels, we adopt binary cross-entropy loss [13] and intersection over union (IoU) loss [34], denoted as:

$$\mathcal{L}_Z = \sum_{i=1}^{5} 2^{1-i} \left(\mathcal{L}_{BCE}(Z_i,\ \tilde{\mathcal{M}}) + \mathcal{L}_{IoU}(Z_i,\ \tilde{\mathcal{M}})\right). \tag{11}$$

In the same way, $\mathcal{L}_{\tilde{Z}}$ can be calculated for the predictions of augmented images. Furthermore, a structural consistency loss [50] is adopted to ensure the prediction consistency over different images.

$$\begin{aligned} \mathrm{SM}\left(\mathrm{p}_1, \mathrm{p}_2\right) &= \tfrac{1-\mathrm{SSIM}(\mathrm{p}_1, \mathrm{p}_2)}{2} \\ \mathcal{L}_{sc} &= \tfrac{1}{n}\textstyle\sum_i (1-\beta)\cdot \mathrm{SM}(Z_{1_i}, \tilde{Z}_{1_i}) + \beta \left| Z_{1_i} - \tilde{Z}_{1_i} \right| \end{aligned}, \tag{12}$$

where SSIM is structural similarity [14], p_1 and p_2 are two pixels. n is the total number of pixels, and i is the index of each pixel. β is set as 0.85. Note that the structure consistency loss is only calculated on final predictions (Z_1 and $\tilde{Z}_1$). Thus, the overall loss is: $\mathcal{L}_{total} = \mathcal{L}_Z + \mathcal{L}_{\tilde{Z}} + \mathcal{L}_{sc}$.

4 Experiments

4.1 Experimental Setup

Dataset. We test our method on three widely used COD benchmarks: CAMO [24], COD10K [11], and NC4K [32]. We adopt the scribble-annotated dataset S-COD [17] for model training.

Evaluation Metrics. We adopt four widely used evaluation metrics, including structure-measure ($\mathbf{S}_\alpha$) [10], mean absolute error ($\mathbf{M}$) [12], adaptive F-measure ($\mathbf{F}_\beta$) [33], and mean E-measure ($\mathbf{E}_\phi$) [9], to evaluate the performance of the proposed method.

Implementation Details. The model is implemented using PyTorch. We utilized SAM-ViT-H [23] as the pre-trained model. For the SAM-based pseudo-label generation, we conduct 10 iterations for random sampling, with about 20 points from foreground and background, respectively, in each iteration. The pre-trained PVT-v2-b4 [43] model trained on ImageNet is adopted as the backbone. During training, Adam is used as the optimizer with a learning rate of 1e-5 for the encoder and 1e-4 for the decoder. The images are resized to 384×384 for input. The complete training process spent 60 epochs with a batch size of 4. It took about 10 hours on a machine with a single NVIDIA 4090 GPU with 24GB of memory. Note that we perform three types of image augmentation: (a) SAM-based pseudo-label generation (Fig. 2), (b) standard data augmentation (*e.g.*, random flips) for model training, and (c) the dual-input data augmentation used to construct structural consistency loss, as shown in Fig. 1.

Table 1: Quantitative comparison of our proposed method and state-of-the-art methods on three benchmark datasets. ↑ (or ↓) denotes the higher (or the lower) the better. The best three results are marked in red , green and blue, respectively. "*" indicates a CRNet variant that is integrated with the PVT-v2 encoder and our pseudo-label generation framework.

Method	CAMO				COD10K				NC4K			
	$\mathbf{M}\downarrow$	$\mathbf{F}_\beta\uparrow$	$\mathbf{E}_\phi\uparrow$	$\mathbf{S}_\alpha\uparrow$	$\mathbf{M}\downarrow$	$\mathbf{F}_\beta\uparrow$	$\mathbf{E}_\phi\uparrow$	$\mathbf{S}_\alpha\uparrow$	$\mathbf{M}\downarrow$	$\mathbf{F}_\beta\uparrow$	$\mathbf{E}_\phi\uparrow$	$\mathbf{S}_\alpha\uparrow$
Fully-Supervised Camouflaged Object Detection												
$SINet_{20}$	.100	.709	.771	.751	.051	.593	.806	.771	.058	.758	.871	.808
$TINet_{21}$	.087	.678	.847	.781	.043	.635	.848	.793	.055	.766	.879	.829
$PFNet_{21}$	.085	.751	.842	.782	.040	.676	.877	.800	.053	.779	.888	.829
$UJSC_{21}$	.073	.779	.859	.800	.035	.705	.884	.809	.047	.803	.898	.842
$UGTR_{21}$	.086	.748	.822	.784	.036	.671	.853	.817	.052	.778	.875	.839
$C^2F\text{-}Net\text{-}v2_{22}$	.077	.763	.842	.790	.034	.731	.880	.813	.050	.805	.887	.834
$SINet\text{-}v2_{22}$	.070	.779	.881	.820	.037	.682	.887	.815	.048	.792	.902	.847
$SegMaR_{22}$	.071	.795	.874	.815	.034	.739	.899	.833	.046	.821	.896	.841
$ZoomNet_{22}$	.066	.792	.877	.820	.029	.741	.888	.838	.043	.814	.896	.853
$DGNet_{23}$	.057	.803	.900	.838	.033	.698	.895	.822	.042	.802	.911	.857
$FEDER_{23}$	.071	.786	.867	.802	.032	.740	.899	.822	.044	.822	.907	.847
EVP_{23}	.059	.808	.897	.846	.029	.736	.908	.844	.039	.822	.918	.874
$FSPNet_{23}$	.050	.829	.899	.856	.026	.736	.895	.851	.035	.826	.915	.879
$SAM\text{-}Adapter_{23}$	.070	.818	.873	.847	.025	.824	.918	.883	.042	.841	.904	.884
$DSAM_{24}$	.061	.824	.915	.832	.033	.759	.921	.846	.040	.844	.932	.871
$CamoFormer_{24}$	.045	.841	.923	.873	.024	.784	.925	.861	.031	.843	.931	.887
CDP_{25}	.072	-	.844	.798	.036	-	.863	.791	.051	-	.902	.880
$DSNet_{25}$	.073	-	.870	.817	.038	-	.878	.809	.050	-	.894	.843
$MAM\text{-}FFC_{25}$	.045	.856	.929	.872	.023	.789	.933	.871	.031	.860	.937	.890
MCF_{25}	.050	-	.915	.864	.023	-	.933	.872	.033	-	.932	.885
$FLRNet_{25}$	.045	-	.927	.877	.024	-	.929	.868	.032	-	.937	.884
Ours	.045	.836	.927	.874	.021	.794	.934	.870	.029	.848	.938	.890
Weakly-Supervised Camouflaged Object Detection												
SAM_{23}	.160	.597	.639	.643	.093	.673	.737	.730	.118	.675	.723	.717
$SAM\text{-}S_{23}$	.105	.682	.774	.731	.046	.695	.828	.772	.071	.747	.832	.763
$WSSA_{20}$	.118	.625	.786	.696	.071	.536	.770	.684	.091	.657	.779	.761
$SCWS_{21}$	.102	.658	.795	.713	.055	.602	.805	.710	.073	.723	.814	.784
TEL_{22}	.104	.681	.797	.717	.057	.633	.826	.724	.075	.754	.832	.782
$CRNet_{23}$	.092	.709	.815	.735	.049	.637	.832	.733	.064	.751	.853	.779
$WS\text{-}SAM_{23}$	.092	**.742**	.818	.759	.038	**.719**	.878	.803	.052	**.802**	.886	.829
$MiNet_{24}$	.091	-	.840	.750	.049	-	.840	.749	.061	-	.869	.793
$SAM\text{-}COD_{24}$	**.060**	-	**.903**	**.836**	**.029**	-	**.904**	**.833**	**.039**	-	**.912**	**.859**
$CRNet\text{-}PVT^*$	**.064**	**.811**	**.889**	**.835**	**.030**	**.738**	**.894**	**.834**	**.044**	**.819**	**.902**	**.855**
Ours	**.055**	**.828**	**.911**	**.849**	**.029**	**.768**	**.910**	**.838**	**.039**	**.836**	**.918**	**.862**

Table 2: Ablation study of proposed prompt enhancement on COD10K. DGS means density-based grid sampling and AUG means image augmentation.

Settings	$\mathbf{M}\downarrow$	$\mathbf{F}_\beta\uparrow$	$\mathbf{E}_\phi\uparrow$	$\mathbf{S}_\alpha\uparrow$
WS-SAM	0.038	0.719	0.878	0.803
w/o AUG	0.030	0.749	0.902	0.830
w/o DGS	0.030	0.753	0.904	0.833
Ours	**0.029**	**0.768**	**0.910**	**0.838**

Table 3: Ablation study of proposed components on COD10K. "SCL" is the structural consistency loss. "AL" is the auxiliary loss for auxiliary outputs.

#	Encoder	Decoder			Loss Setting		COD10K			
	PVT-v2	CRNet	ViT Block	CSC	SCL	AL	$\mathbf{M}\downarrow$	$\mathbf{F}_\beta\uparrow$	$\mathbf{E}_\phi\uparrow$	$\mathbf{S}_\alpha\uparrow$
①	✓	✓			✓	✓	0.030	0.739	0.894	0.834
②	✓		✓		✓	✓	0.031	0.731	0.899	0.833
③	✓			✓			0.031	0.756	0.889	0.822
④	✓			✓		✓	0.029	**0.769**	0.899	0.833
⑤	✓			✓	✓		0.030	0.745	0.904	0.834
⑥	✓			✓	✓	✓	**0.029**	0.768	**0.910**	**0.838**

4.2 *Comparison with State-of-the-arts*

Competitors. Our proposed method is compared with 31 recent state-of-the-art methods, including 1) 12 CNN-based fully-supervised methods, such as SINet [11], TINet [58], PFNet [35], UJSC [26], C^2F-Net-v2 [2], SINet-v2 [12], SegMaR [22], ZoomNet [37], FEDER [15], DSNet [47], CDP [31] and MCF [40]; 2) 9 Transformer-based fully-supervised methods, such as UGTR [46], CamoFormer [49], FSPNet [19], DGNet [20], EVP [30], SAM-Adapter [5], DSAM [51], MAM-FFC [42] and FLRNet [56]; and 3) 7 weakly-supervised methods, such as WSSA [53], SCWS [50], TEL [29], CRNet [17], WS-SAM [16], MiNet [36] and SAM-COD [3]. Besides, we also introduce two SAM methods for comparison, one of which applies SAM directly on the test, and the other SAM-S [16] fine-tunes SAM with scribble annotations. For comparison, we also introduce a variant of CRNet that employs the same PVT-v2-b4 backbone and adopts our pseudo-label generation strategy for prediction supervision.

Qualitative Evaluation. Fig. 5 shows visual comparisons across several challenging scenarios, including single-object, multi-object, large-object, and small-object cases. Our method demonstrates robust performance, producing more accurate and complete predictions than existing approaches, including some fully supervised methods.

Quantitative Evaluation. Tab. 1 shows quantitative comparisons on three COD benchmarks. Our method achieves the best performance among weakly supervised methods, showing significant improvements over recent models such as MiNet and SAM-COD. Notably, it even surpasses several outstanding fully supervised models, including ZoomNet, DGNet, and FEDER. To further validate the effectiveness, we integrate the proposed pseudo-label generation strategy and transformer backbone into the recently proposed CRNet, resulting in a variant referred to as CRNet-PVT. As shown in the results, CRNet-PVT exhibits marked performance gains, demonstrating the benefit of our model components. Nonetheless, our method still outperforms this variant, highlighting the advantages of incorporating SAM-based pseudo-labeling and transformer architecture.

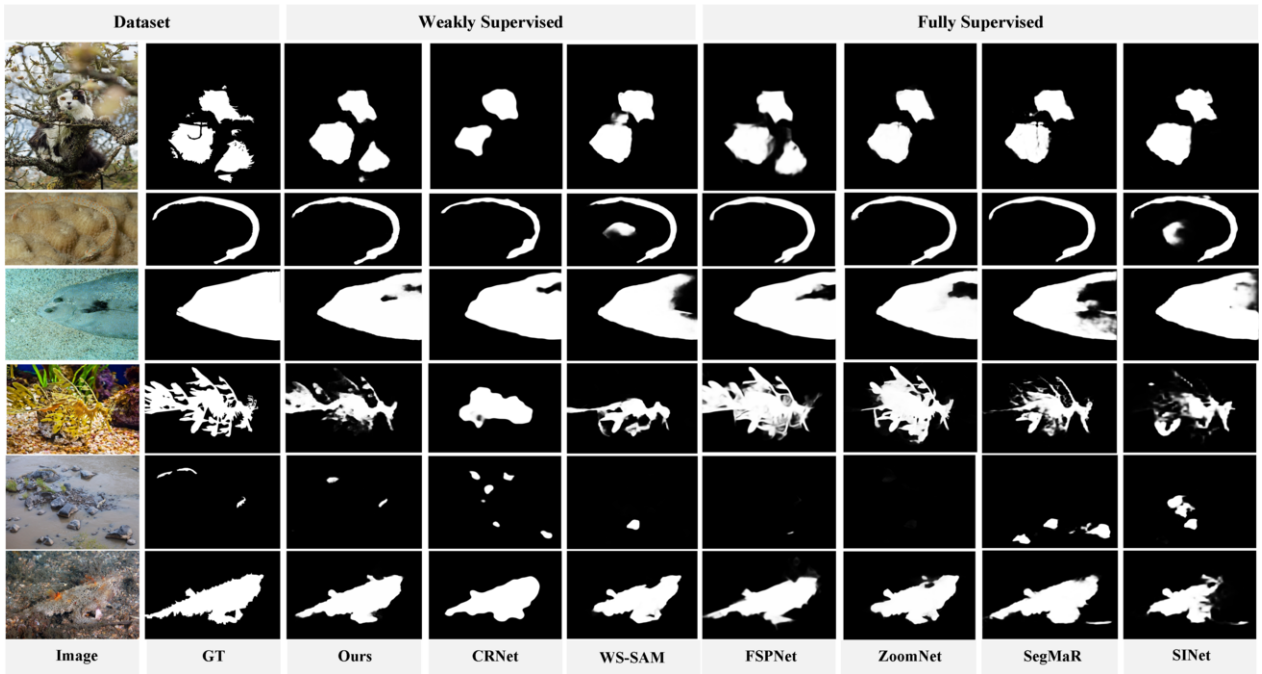


Figure 5: Visual comparison of our proposed method and other representative COD methods.

			CAMO				COD10K				NC4K			
$\mathcal{K}$	$\mathcal{P}$	τ	$\mathbf{M}\downarrow$	$\mathbf{F}_\beta\uparrow$	$\mathbf{E}_\phi\uparrow$	$\mathbf{S}_\alpha\uparrow$	$\mathbf{M}\downarrow$	$\mathbf{F}_\beta\uparrow$	$\mathbf{E}_\phi\uparrow$	$\mathbf{S}_\alpha\uparrow$	$\mathbf{M}\downarrow$	$\mathbf{F}_\beta\uparrow$	$\mathbf{E}_\phi\uparrow$	$\mathbf{S}_\alpha\uparrow$
10	20	1	.070	.819	.868	.811	.033	.762	.888	.813	.052	.833	.885	.829
10	20	2	.061	.836	.892	.831	.030	.782	.904	.831	.044	.840	.904	.847
10	20	3	.056	.841	.904	.847	.029	.774	.906	.835	.042	.840	.907	.855
10	**20**	**4**	.055	.828	.911	.849	.029	.768	.910	.838	.039	.836	.918	.862
5	20	4	.056	.825	.908	.851	.029	.767	.903	.834	.041	.836	.908	.856
10	**20**	**4**	.055	.828	.911	.849	.029	.768	.910	.838	.039	.836	.918	.862
15	20	4	.057	.806	.905	.850	.030	.737	.903	.830	.040	.821	.915	.861
20	20	4	.055	.814	.909	.853	.030	.745	.902	.829	.040	.826	.914	.861
10	10	4	.057	.822	.901	.837	.030	.748	.900	.827	.042	.824	.907	.854
10	**20**	**4**	.055	.828	.911	.849	.029	.768	.910	.838	.039	.836	.918	.862
10	30	4	.055	.821	.906	.849	.029	.749	.904	.830	.041	.827	.912	.859
10	50	4	.055	.834	.906	.853	.029	.762	.907	.837	.041	.831	.913	.862

Table 4: Ablation study on parameter settings of the SAM-based pseudo label generation. The best results are marked in red.

4.3 Ablation Study

Prompt Enhancement. To verify the effectiveness of the proposed prompt enhancement strategy, we made adjustments by removing density-based grid sampling and image augmentation, respectively, for comparison, shown in Tab. 2. It is evident that our method achieves the best performance and improves the quality of SAM-based mask generation (WS-SAM).

Cross-scale Correlation Module. To demonstrate the effectiveness of our cross-scale correlation module, we compare it with the CRNet decoder and standard ViT decoder, as shown in ①, ②, and ⑥ of Tab. 3. We can see that the proposed module achieves significant performance improvement, especially in terms of F_β and E_ϕ.

SAM-based Pseudo Label Generation. We performed ablation experiments on the number of image augmentation $\mathcal{K}$, the number of sample points $\mathcal{P}$, and the threshold τ. As shown in Tab. 4, the setting $\mathcal{K} = 10$, $\mathcal{P} = 20$, $\tau = 4$ achieves the best overall performance.

Loss Function. We test the effectiveness of loss functions, as shown in ③, ④, ⑤, and ⑥ of Tab. 3. With auxiliary loss and structure consistency loss, the model performance has been enhanced to different extents. The combined version yields the best performance.

5 Conclusions

In this paper, we propose a novel SAM-guided multi-scale feature correlation transformer for WSCOD. Firstly, to improve weak supervision, we design a SAM-based label generation module that leverages sparse annotations as prompts of SAM to generate reliable pseudo object masks for model training, under two strategies, *i.e.*, density-based grid sampling for prompt enhancement and label response filter. Then, to capture multi-scale camouflage cues, we propose a novel cross-scale correlation module in the transformer framework, which utilizes enriched cross-attention operations to excavate global context and aggregate multi-scale discriminative features for accurate segmentation. Extensive experiments on three benchmark datasets demonstrate the state-of-the-art performance of our method over existing weakly supervised methods, even surpassing some fully supervised methods.

References

[1] J. S. Chahl and H. Liu. Bioinspired invertebrate pest detection on standing crops. In *Bioinspiration, Biomimetics, and Bioreplication VIII*, volume 10593, pages 68–81. SPIE, 2018.

[2] G. Chen, S.-J. Liu, Y.-J. Sun, G.-P. Ji, Y.-F. Wu, and T. Zhou. Camouflaged object detection via context-aware cross-level fusion. *TCSVT*, 32 (10):6981–6993, 2022.

[3] H. Chen, P. Wei, G. Guo, and S. Gao. SAM-COD: SAM-guided unified framework for weakly-supervised camouflaged object detection. In *ECCV*, 2024.

[4] T. Chen, Z. Mai, R. Li, and W.-L. Chao. Segment anything model (SAM) enhanced pseudo labels for weakly supervised semantic segmentation. In *NeurIPS Workshop*, 2023.

[5] T. Chen, L. Zhu, C. Deng, R. Cao, Y. Wang, S. Zhang, Z. Li, L. Sun, Y. Zang, and P. Mao. Sam-adapter: Adapting segment anything in underperformed scenes. In *ICCVW*, pages 3367–3375, 2023.

[6] Z. Chen, R. Gao, T.-Z. Xiang, and F. Lin. Diffusion model for camouflaged object detection. In *ECAI*, volume 372, pages 445–452, 2023.

[7] Z. Chen, K. Sun, X. Lin, and R. Ji. Camodiffusion: Camouflaged object detection via conditional diffusion models. In *AAAI*, volume 38, pages 1272–1280, 2023.

[8] B. Dong, J. Pei, R. Gao, T.-Z. Xiang, S. Wang, and H. Xiong. A unified query-based paradigm for camouflaged instance segmentation. In *ACM MM*, pages 2131–2138, 2023.

[9] D. Fan, C. Gong, Y. Cao, B. Ren, M. Cheng, and A. Borji. Enhanced-alignment measure for binary foreground map evaluation. In *IJCAI*, pages 698–704, 2018.

[10] D.-P. Fan, M.-M. Cheng, Y. Liu, and etc. Structure-measure: A new way to evaluate foreground maps. In *ICCV*, pages 4548–4557, 2017.

[11] D.-P. Fan, G.-P. Ji, G. Sun, M.-M. Cheng, J. Shen, and L. Shao. Camouflaged object detection. In *CVPR*, pages 2777–2787, 2020.

[12] D.-P. Fan, G.-P. Ji, M.-M. Cheng, and L. Shao. Concealed object detection. *IEEE TPAMI*, 44(10):6024–6042, 2022.

[13] D.-P. Fan, G.-P. Ji, P. Xu, M.-M. Cheng, C. Sakaridis, and L. Van Gool. Advances in deep concealed scene understanding. *VI*, 1(1):16, 2023.

[14] C. Godard, O. Mac Aodha, and G. J. Brostow. Unsupervised monocular depth estimation with left-right consistency. In *CVPR*, 2017.

[15] C. He, K. Li, Y. Zhang, L. Tang, Y. Zhang, Z. Guo, and X. Li. Camouflaged object detection with feature decomposition and edge reconstruction. In *CVPR*, pages 22046–22055, 2023.

[16] C. He, K. Li, Y. Zhang, G. Xu, L. Tang, Y. Zhang, Z. Guo, and X. Li. Weakly-supervised concealed object segmentation with sam-based pseudo labeling and multi-scale feature grouping. *NeurIPS*, 2023.

[17] R. He, Q. Dong, J. Lin, and etc. Weakly-supervised camouflaged object detection with scribble annotations. In *AAAI*, pages 781–789, 2023.

[18] X. Hu, S. Wang, X. Qin, H. Dai, W. Ren, D. Luo, Y. Tai, and L. Shao. High-resolution iterative feedback network for camouflaged object detection. In *AAAI*, pages 881–889, 2023.

[19] Z. Huang, H. Dai, T.-Z. Xiang, S. Wang, H.-X. Chen, J. Qin, and H. Xiong. Feature shrinkage pyramid for camouflaged object detection with transformers. In *CVPR*, pages 5557–5566, 2023.

[20] G.-P. Ji, D.-P. Fan, Y.-C. Chou, D. Dai, A. Liniger, and L. Van Gool. Deep gradient learning for efficient camouflaged object detection. *MIR*, 20(1):92–108, 2023.

[21] G.-P. Ji, D.-P. Fan, P. Xu, M.-M. Cheng, B. Zhou, and L. Van Gool. Sam struggles in concealed scenes–empirical study on" segment anything". *SCIS*, 66(226101), 2023.

[22] Q. Jia, S. Yao, Y. Liu, X. Fan, R. Liu, and Z. Luo. Segment, magnify and reiterate: Detecting camouflaged objects the hard way. In *CVPR*, pages 4713–4722, 2022.

[23] A. Kirillov, E. Mintun, N. Ravi, H. Mao, C. Rolland, L. Gustafson, T. Xiao, S. Whitehead, A. C. Berg, W.-Y. Lo, P. Dollar, and R. Girshick. Segment anything. In *ICCV*, pages 4015–4026, 2023.

[24] T.-N. Le, T. V. Nguyen, Z. Nie, M.-T. Tran, and A. Sugimoto. Anabranch network for camouflaged object segmentation. *CVIU*, 184: 45–56, 2019.

[25] C. Lei, J. Fan, X. Li, T. Xiang, A. Li, C. Zhu, and L. Zhang. Towards real zero-shot camouflaged object segmentation without camouflaged annotations. *IEEE TPAMI*, pages 1–1, 2025.

[26] A. Li, J. Zhang, Y. Lv, B. Liu, T. Zhang, and Y. Dai. Uncertainty-aware joint salient object and camouflaged object detection. In *CVPR*, pages 10071–10081, 2021.

[27] L. Li, J. Liu, F. Yu, X. Wang, and T.-Z. Xiang. MVDI25K: A large-scale dataset of microscopic vaginal discharge images. *TBench*, 1(1), 2021.

[28] L. Li, J. Liu, S. Wang, X. Wang, and T.-Z. Xiang. Trichomonas vaginalis segmentation in microscope images. In *MICCAI*, pages 68–78, 2022.

[29] Z. Liang, T. Wang, X. Zhang, J. Sun, and J. Shen. Tree energy loss: Towards sparsely annotated semantic segmentation. In *CVPR*, pages 16907–16916, 2022.

[30] W. Liu, X. Shen, C.-M. Pun, and etc. Explicit visual prompting for low-level structure segmentations. In *CVPR*, pages 19434–19445, 2023.

[31] Y. Liu, C. Li, X. Dong, L. Li, D. Zhang, S. Xu, and J. Han. Seamless detection: Unifying salient object detection and camouflaged object detection. *ESWA*, 274:126912, 2025.

[32] Y. Lv, J. Zhang, Y. Dai, A. Li, B. Liu, N. Barnes, and D.-P. Fan. Simultaneously localize, segment and rank the camouflaged objects. In *CVPR*, pages 11591–11601, 2021.

[33] R. Margolin, L. Zelnik-Manor, and A. Tal. How to evaluate foreground maps? In *CVPR*, pages 248–255, 2014.

[34] G. Máttyus, W. Luo, and R. Urtasun. Deeproadmapper: Extracting road topology from aerial images. In *ICCV*, pages 3438–3446, 2017.

[35] H. Mei, G.-P. Ji, Z. Wei, X. Yang, X. Wei, and D.-P. Fan. Camouflaged object segmentation with distraction mining. In *CVPR*, pages 8772–8781, 2021.

[36] Y. Niu, L. Yang, R. Xu, Y. Li, and Y. Chen. Minet: Weakly-supervised camouflaged object detection through mutual interaction between region and edge cues. In *ACM MM*, page 6316–6325, 2024.

[37] Y. Pang, X. Zhao, T.-Z. Xiang, L. Zhang, and H. Lu. Zoom in and out: A mixed-scale triplet network for camouflaged object detection. In *CVPR*, pages 2160–2170, 2022.

[38] Y. Pang, X. Zhao, T.-Z. Xiang, L. Zhang, and H. Lu. Zoomnext: A unified collaborative pyramid network for camouflaged object detection. *IEEE TPAMI*, 46(12):9205–9220, 2024.

[39] R. Pérez-de la Fuente, X. Delclòs, E. Peñalver, M. Speranza, J. Wierzchos, C. Ascaso, and M. S. Engel. Early evolution and ecology of camouflage in insects. *PNAS*, 109(52):21414–21419, 2012.

[40] W. Sun, Q. Wang, Y. Tian, X. Yang, X. Kong, Y. Dong, and Y. Zhang. Multi-level cross-knowledge fusion with edge guidance for camouflaged object detection. *KBS*, page 113070, 2025.

[41] Y. Sun, S. Wang, C. Chen, and T.-Z. Xiang. Boundary-guided camouflaged object detection. In *IJCAI*, pages 1335–1341, 2022.

[42] T. Wang, Z. Yu, J. Fang, J. Xie, F. Yang, H. Zhang, L. Zhang, M. Du, L. Li, and X. Ning. Multidimensional fusion of frequency and spatial domain information for enhanced camouflaged object detection. *Information Fusion*, 117:102871, 2025.

[43] W. Wang, E. Xie, X. Li, D.-P. Fan, K. Song, D. Liang, T. Lu, P. Luo, and L. Shao. Pvt v2: Improved baselines with pyramid vision transformer. *CVM*, 8(3):415–424, 2022.

[44] R. Wu, T.-Z. Xiang, G.-S. Xie, R. Gao, X. Shu, F. Zhao, and L. Shao. Uncertainty-aware transformer for referring camouflaged object detection. *IEEE TIP*, pages 1–1, 2025.

[45] Z. Wu, D. P. Paudel, D.-P. Fan, J. Wang, S. Wang, C. Demonceaux, R. Timofte, and L. Van Gool. Source-free depth for object pop-out. In *ICCV*, pages 1032–1042, 2023.

[46] F. Yang, Q. Zhai, X. Li, R. Huang, A. Luo, H. Cheng, and D.-P. Fan. Uncertainty-guided transformer reasoning for camouflaged object detection. In *ICCV*, pages 4146–4155, 2021.

[47] H. Yang, Y. Zhu, K. Sun, H. Ding, and X. Lin. Camouflaged object detection via dual-branch fusion and dual self-similarity constraints. *PR*, 157:110895, 2025.

[48] S. Yao, H. Sun, T.-Z. Xiang, X. Wang, and X. Cao. Hierarchical graph interaction transformer with dynamic token clustering for camouflaged object detection. *IEEE TIP*, 33:5936–5948, 2024.

[49] B. Yin, X. Zhang, D.-P. Fan, S. Jiao, M.-M. Cheng, L. Van Gool, and Q. Hou. Camoformer: Masked separable attention for camouflaged object detection. *IEEE TPAMI*, 46(12):10362–10374, 2024.

[50] S. Yu, B. Zhang, J. Xiao, and E. G. Lim. Structure-consistent weakly supervised salient object detection with local saliency coherence. In *AAAI*, pages 3234–3242, 2021.

[51] Z. Yu, X. Zhang, L. Zhao, Y. Bin, and G. Xiao. Exploring deeper! segment anything model with depth perception for camouflaged object detection. In *ACM MM*, pages 4322–4330, 2024.

[52] C. Zhang, H. Bi, T.-Z. Xiang, R. Wu, J. Tong, and X. Wang. Collaborative camouflaged object detection: A large-scale dataset and benchmark. *IEEE TNNLS*, 2023.

[53] J. Zhang, X. Yu, A. Li, P. Song, B. Liu, and Y. Dai. Weakly-supervised salient object detection via scribble annotations. In *CVPR*, pages 12546–12555, 2020.

[54] Y. Zhang, J. Zhang, W. Hamidouche, and O. Deforges. Predictive uncertainty estimation for camouflaged object detection. *IEEE TIP*, 2023.

[55] H. Zhao, J. Shi, X. Qi, X. Wang, and J. Jia. Pyramid scene parsing network. In *CVPR*, pages 2881–2890, 2017.

[56] Y. Zhao, Q. Zhang, and Y. Li. Flrnet: A bio-inspired three-stage network for camouflaged object detection via filtering, localization and refinement. *Neurocomputing*, page 129523, 2025.

[57] Y. Zhong, B. Li, L. Tang, and etc. Detecting camouflaged object in frequency domain. In *CVPR*, pages 4504–4513, 2022.

[58] J. Zhu, X. Zhang, S. Zhang, and J. Liu. Inferring camouflaged objects by texture-aware interactive guidance network. In *AAAI*, pages 3599–3607, 2021.